

AdaBoost Regression

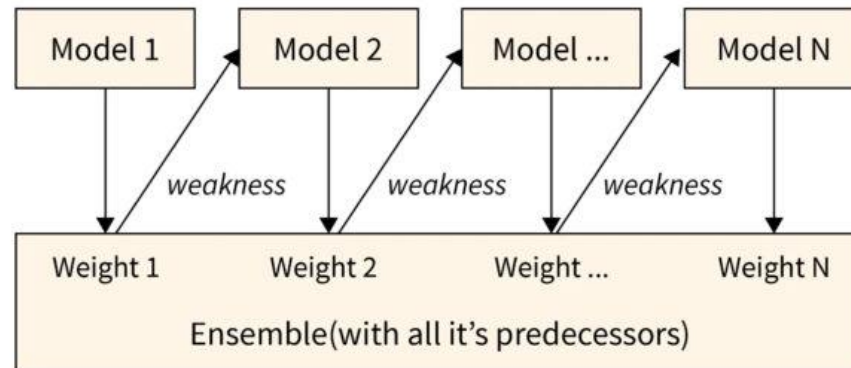
AdaBoost Regression Algorithm

- ▶ The AdaBoost Regressor is an ensemble machine learning algorithm
- ▶ That sequentially builds multiple "weak learners"
- ▶ It evaluates errors from previous models
- ▶ Forcing the next learner to focus on difficult-to-predict values

PROCESS

- ▶ **Initialize Weights:** Every data point in the training set starts with an equal weight.
- ▶ **Train & Predict:** A weak learner (a Decision Tree with $\text{max_depth}=1$) makes predictions.
- ▶ **Evaluate Error:** Calculate the error for each point. Points with higher errors get increased importance/weight.
- ▶ **Sequential Training:** The next model learns specifically from the mistakes of the previous ones

Model 1,2,...,N are individual models (eg. decision tree)



Thank You!